Understanding Log
(Reading Material)

APNA COLLEGE

What is log?

A logarithm is the inverse operation to exponentiation. For a given number x, the logarithm $\log_b(x)$ is the power to which the base b must be raised to obtain x.

Mathematically:

$\log_b(x) = y$

if and only if

$b^y = x$

For example:

$\log_2(8) = 3$

because $2^3 = 8$.

Basic Properties of log

1. **Product Rule:**

   $$\log_b(xy) = \log_b(x) + \log_b(y)$$

2. **Quotient Rule:**

   $$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$$

3. **Power Rule:**

   $$\log_b(x^k) = k \cdot \log_b(x)$$

4. **Change of Base Formula:**

   $$\log_b(x) = \frac{\log_k(x)}{\log_k(b)}$$

   This is useful for converting between different bases.